Unit 7, Lesson 1: Circle Graphs

Benchmark

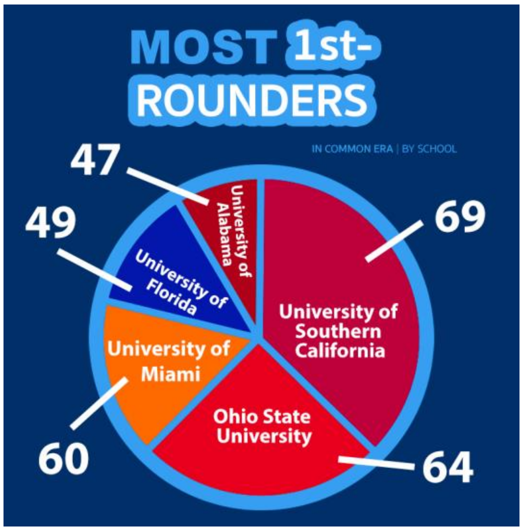
MA.7.DP.1.4 Use proportional reasoning to construct, display, and interpret data in circle graphs.

Learning Target

☐ I can show how an angle measure is related to a portion of a circle.

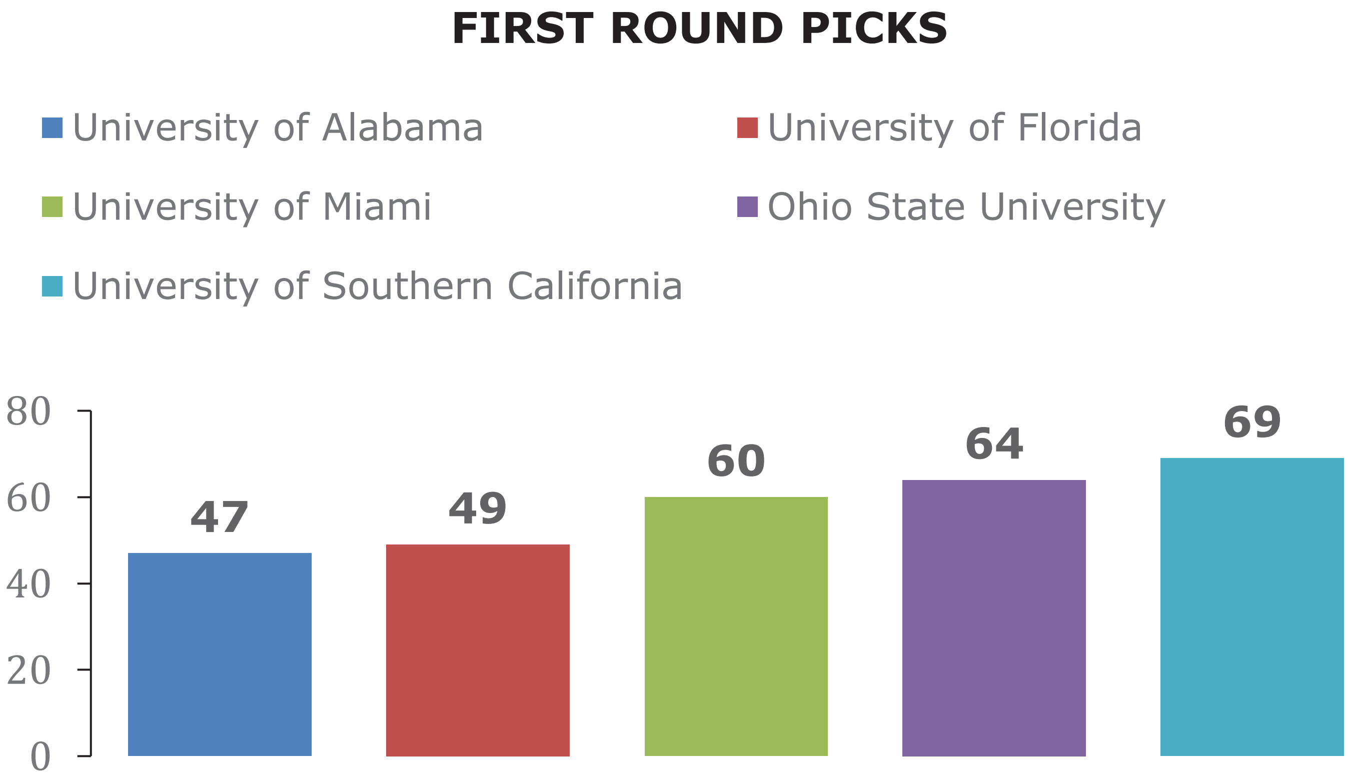
7.1.1 Warm-Up: NFL Draft

The circle graph shows the number of National Football League's (NFL) first round picks for five schools for the common era ending in 2017.



1. What do you notice about the circle graph?

2. A data scientist created a bar chart for the same data.



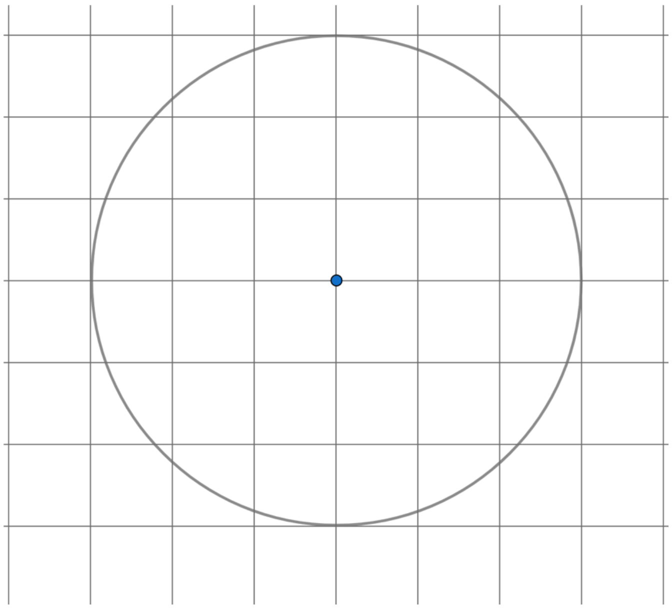
Describe how the bar chart better represents the data than the circle graph.

7.1.2 Exploration: Sectors and Circles

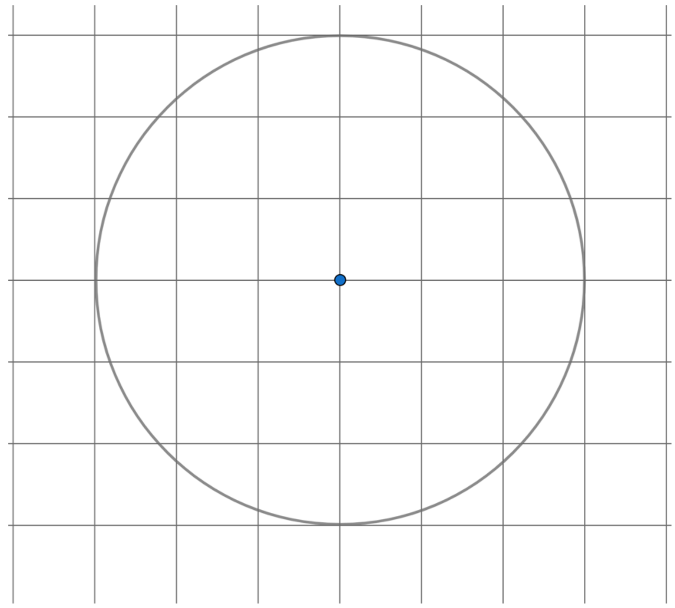
A ***sector*** is a fractional part of the area of a circle.

1. There are six congruent circles drawn below. Use a protractor to divide the circles into sectors that represent halves, thirds, fourths, fifths, sixths, and eighths, as indicated.

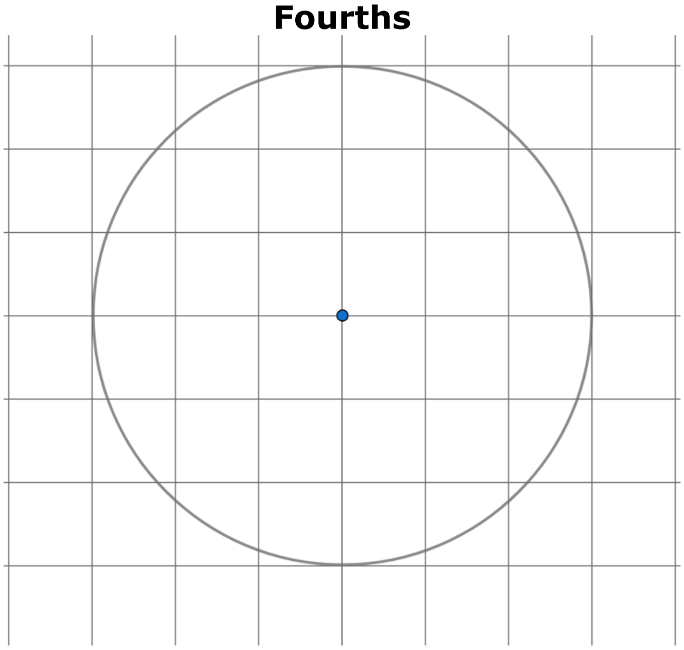
Halves



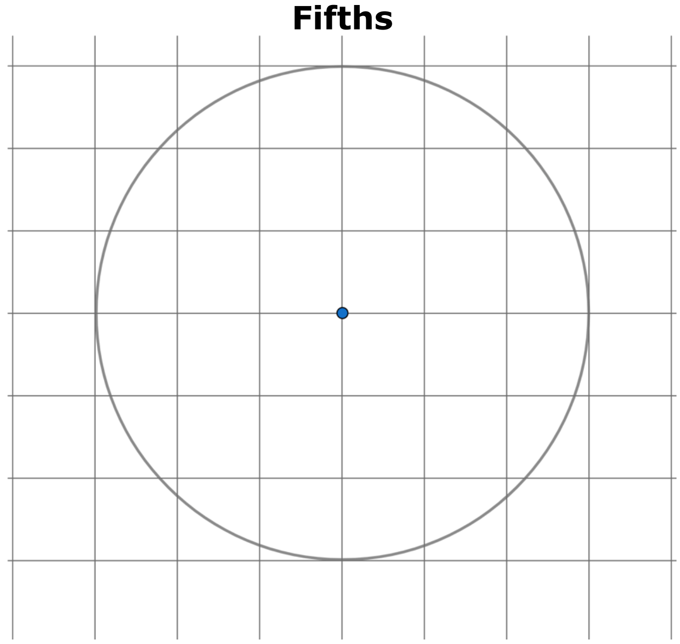
Thirds



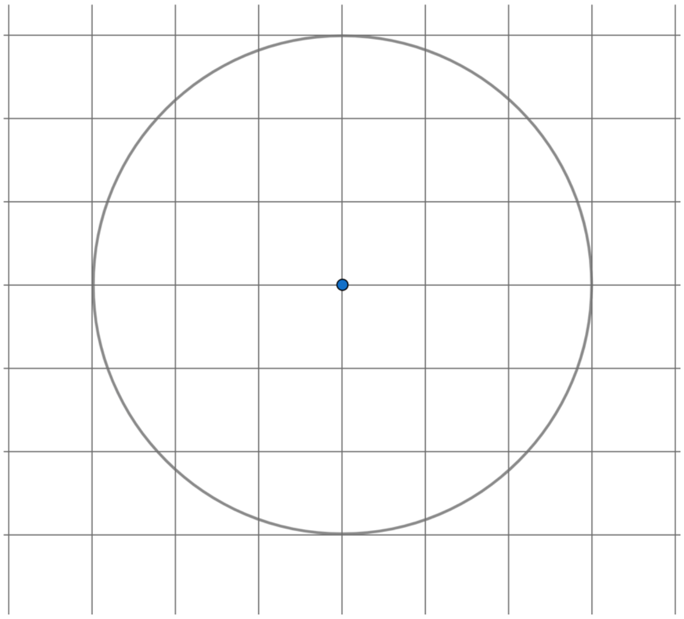
Fourths



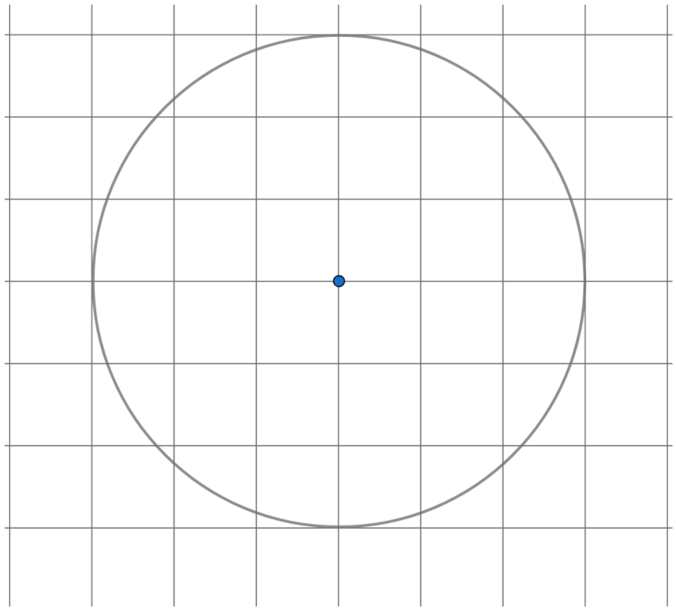
Fifths



Sixths



Eighths



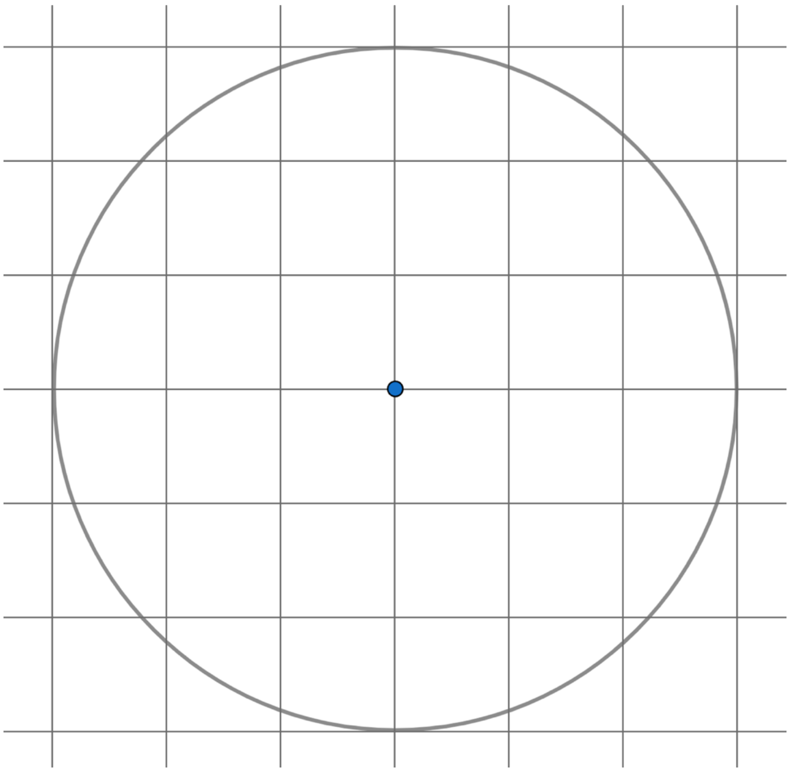
2. Complete the table with the fraction and percentage of the whole circle that each congruent sector represents.

Sector	Fraction	Percentage
Halves		
Thirds		
Fourths		
Fifths		
Sixths		
Eighths		

7.1.3 Collaboration: Dividing Circles into Sectors

In the Exploration, all of the circles were divided into congruent sectors. Data displayed in ***circle graphs*** are not typically divided equally among each category.

1. Use the circle on the coordinate grid to answer the following questions.



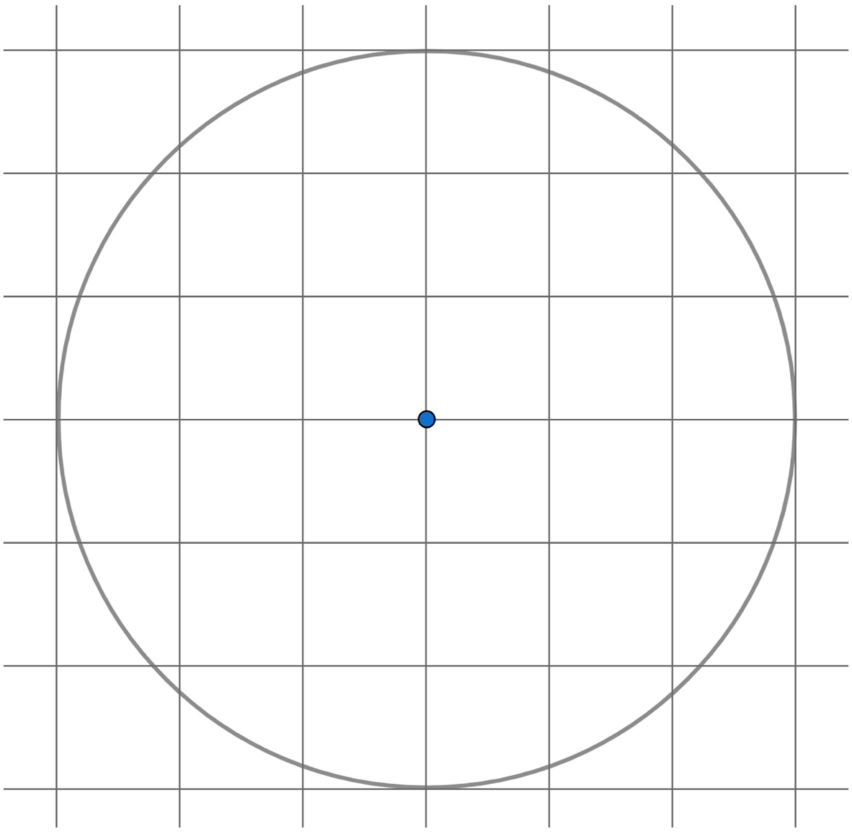
a. Discuss with your partner how to divide the circle into a half and two quarters.

b. Use a straight edge to divide the circle. Label each sector with the fraction and percentage of the whole circle the sector represents.

c. What is the sum of all the fractions in the labeled circle?

d. What is the sum of all the percentages in the labeled circle?

2. Use the circle on the coordinate grid to answer the following questions.



a. Discuss with your partner how to divide the circle into a half, a third, and a sixth.

b. Use a straight edge to divide the circle. Label each sector with the fraction and percentage of the whole circle the sector represents.

c. What is the sum of all the fractions in the labeled circle?

d. What is the sum of all the percentages in the labeled circle?

3. Consider a circle divided into any number of sectors.

a. What should be the sum of all fractions in order to represent the area of the whole circle?

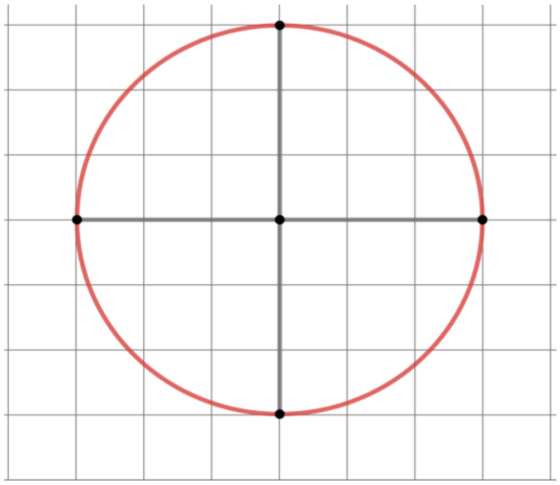
b. What should be the sum of all the percentages in order to represent the area of the whole circle?

7.1.4 Guided Instruction: Determining Central Angle Measures

An ***angle*** is formed wherever two lines, segments, or rays intersect. Angles are measured in degrees.

A ***central angle*** has its vertex at the center of a circle with radii as its sides.

A diagram of a circle with four congruent sectors is shown.



Use the definition of a central angle to determine how many central angles are in the circle with four congruent sectors.

1. Complete the statements.

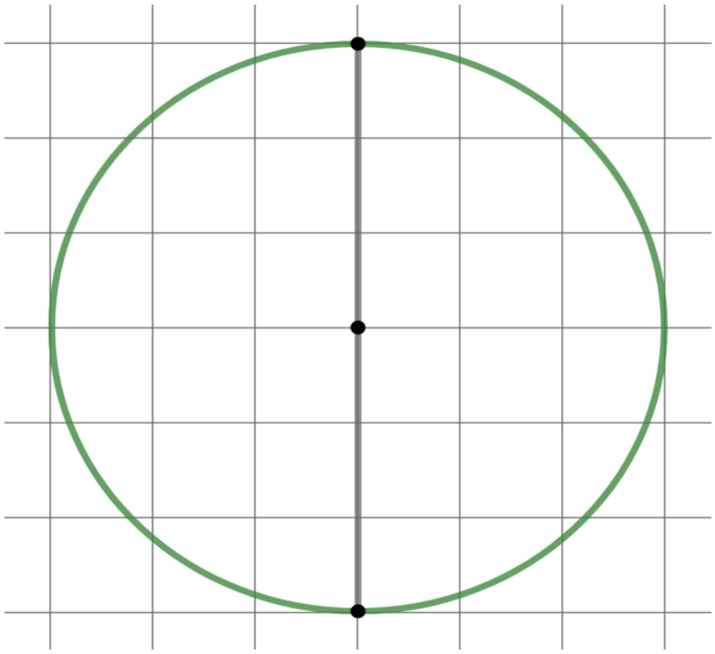
Each of the equal sectors form a [○ right / ○ straight] angle in the center of the circle. A [○ right / ○ straight] angle is a 90 degree angle.

The total sum of the 4 central angles equals [○ 90 degrees. / ○ 360 degrees.]

The measure of each central angle can be written as a ratio of the angle to $360°$. Thus, each sector in the circle divided into fourths takes up $\frac{90}{360}$ of the degrees of the circle.

2. How does this ratio compare to the fraction used to label the sector in the Exploration?

3. A circle that is divided into halves is shown.



a. What is the measure of one of the central angles?

b. What is the name of the type of angle that represents the central angle?

c. Write the ratio of the measure of the central angle to $360°$.

d. How does this ratio compare to the fraction used to label the sector in the Exploration?

4. Answer the following with your partner.

a. If a circle is divided into any number of congruent sectors, how can the central angle measures be calculated? Summarize your discussion.

b. Based on your discussion, fill in the table.

Circle Division	**Central Angle Measure**	**Ratio of Angle Measure to $360°$**
Thirds		
Fifths		
Sixths		
Eighths		

c. Compare the ratios in the table to the fractions used to label the circles in the Exploration. Discuss what you notice with your partner.

d. Write a conjecture of how the angle measure ratio relates to the amount of a circle.

7.1.5 Wrap-Up: Reflection

Complete each statement based on your understanding of today's lesson.

1. I am still unsure about ____

   ____

   _______________________.

2. One question I have that will help me better understand is

   ____

   ____

   ______________________?